Sean Jackson

Website | LinkedIn | GitHub

sjackson6325@gmail.com | 201-937-9681

Citizenship: US / UK / CA

Education

- **University of Toronto** — Toronto, ON M5S 1A1
  Honours BSc. Physics; Minors in Computer Science & Mathematics — *September. 2022 – May. 2026*
  - Clubs: AI Club, Quantum Computing Club, Physics Student Union, Computer Science Student Union

Work Experience

- **Lawrence Livermore National Laboratory** — Livermore, CA 94550, United States
  Machine Learning Engineer (Materials Engineering Division, Summer Intern) — *May. 2026 – Present*
  - **Materials Engineering**: Working within the Materials Engineering Division (MED) on machine learning applications for advanced materials research and manufacturing.
- **Lawrence Livermore National Laboratory** — Livermore, CA 94550, United States
  Machine Learning Engineer (Data Science Scholar Institute, Summer Intern) — *May. 2025 – September. 2025*
  - **Project APEX**: Designed, trained, and deployed deep computer-vision models for high-throughput materials characterization in a self-driving laboratory for automated alloy discovery. Curated training datasets with Cornell University.
  - **Systems Design**: Built low-level robotics and tooling frameworks, integrated into high-level architectures for closed-loop automation (*details restricted under U.S. DOE regulations*).
  - **CV Design Challenge**: Designed generative decoding models for tracking occluded objects through walls using temporal feature representations.

Publications

- **Classifying Alloy Surface Preparation Quality with Metadata Infused Machine Learning for Rapid Alloy Discovery**: Under Review (2026). Novel end-to-end CV framework for automated materials assessment in self-driving labs. Led model design, implementation, and evaluation; imaging data from Cornell, validation with LLNL.

Projects - Github

- **GRACE — GRinding Automated Classification Engine** — GitHub
  Custom PyTorch CV framework for automated materials quality assessment in self-driving labs. First-author paper under review.
- **SmartQueue** — GitHub | Website
  UofT office hours queue system with dynamic wait-time estimation, thematic clustering (DBScan + SBERT), and hybrid in-person/remote support.
- **Time Management Manager (TMM)** — GitHub
  Group time management system with dynamic multi-user availability and automated task scheduling.

Skills

- **ML & AI**: PyTorch, computer vision, architecture design, dataset engineering, model deployment (edge/limited-compute), RAG, LLMs, Transformers.
- **Software & Infrastructure**: Python, C/C++, Java, Linux, SQL, Git, CI/CD (GitHub Actions, Jenkins), AWS, Docker, Arduino/Raspberry Pi, MATLAB.
- **High Performance Computing**: Trained ML models and executed large-scale simulations on LLNL supercomputing clusters, including Sierra and Lassen.
- **Physics & Instrumentation**: Computational physics (PDEs, Monte Carlo, FFT), laboratory instrumentation (lasers, optics, oscilloscopes, lock-in amplifiers), semiconductor fabrication (sputtering, photolithography).
- **Quantum Computing**: Qiskit, Cirq, quantum algorithms, quantum information theory, quantum key encryption.